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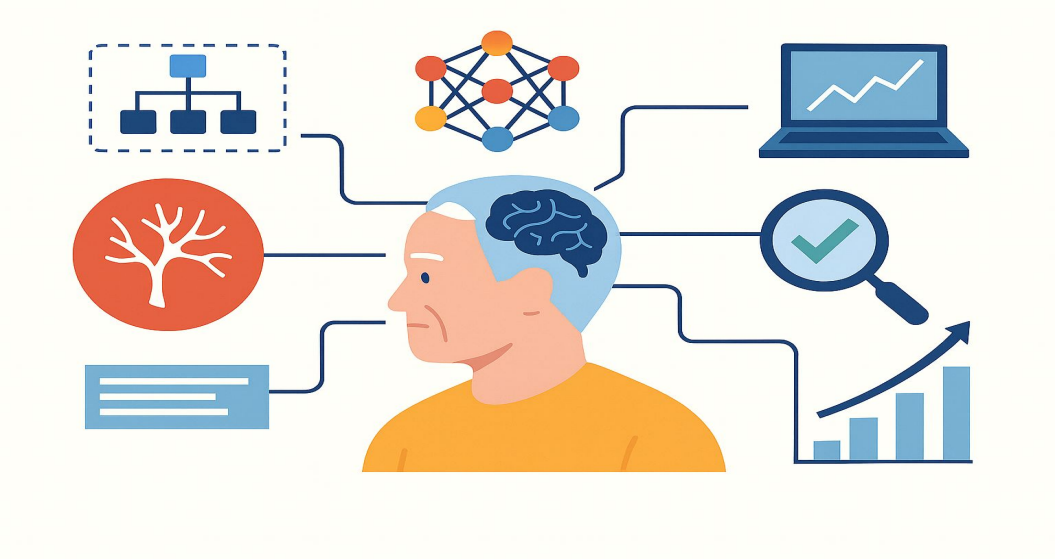
Prediction of Postoperative Stroke in Elderly Patients with CATBoost

Group 10:

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- Abhijeet Nijjar
- Himaja Kavuri

OUTLINE

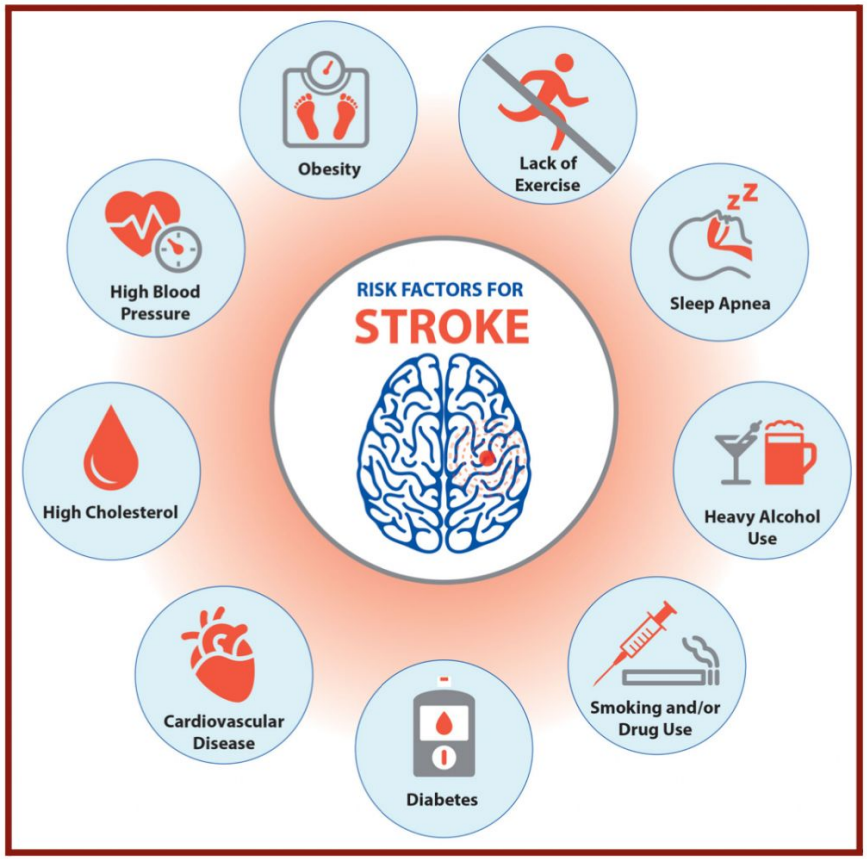
- Objective & Background
- Overview
- Literature Review
- Data Source
- Data Extraction
- Process Pipeline
- Data Preprocessing
- Feature Selection
- Data Modeling
- Results
- Proposed Model
- Statistical tests
- Model Interpretation
- Conclusion
- Citations



Objective

Objective

To develop a CATBoost-based predictive model capable of accurately identifying elderly patients that are at a high risk of postoperative stroke.



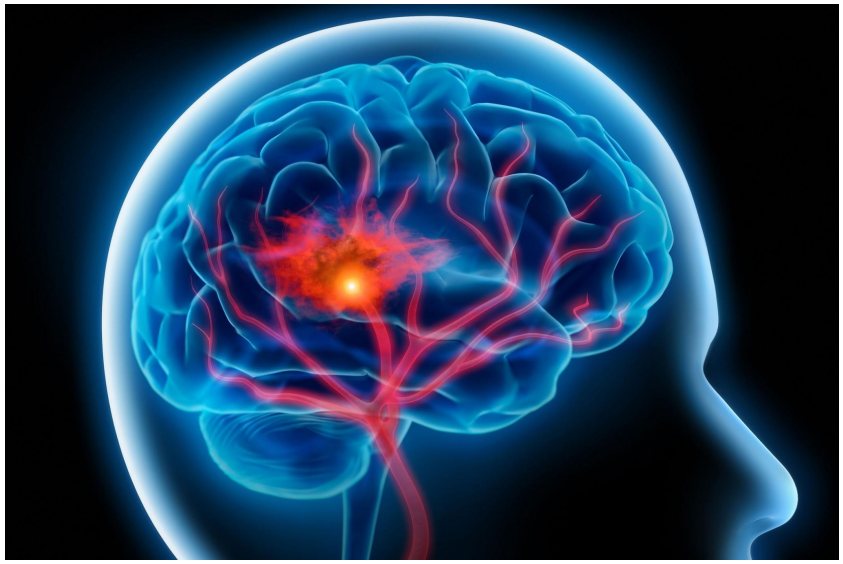
Background:

- Postoperative stroke is a rare but life-threatening complication, especially in elderly surgical patients.
- Early identification of high-risk patients is difficult because stroke arises from a complex interplay of comorbidities, metabolic changes, and pre-operative factors.
- Traditional risk assessment tools often lack predictive accuracy or are not individualized for postoperative dynamics.
- Machine learning techniques can capture nonlinear relationships and feature interactions that classical statistical models may miss.

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OVERVIEW

- Postoperative stroke is a severe complication caused by disrupted cerebral blood flow after surgery.
- Elderly patients are at particularly high risk due to frailty and comorbidities.
- It carries a high burden of mortality, disability, and prolongs the ICU stay.
- Early prediction is crucial but challenging because risk factors are multifactorial and nonlinear.



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LITERATURE REVIEW

Study	Proposed Method	Performance
Zhang et al. (2022)	XGBoost	AUROC:0.83 95% CI: (0.79 - 0.87)

- Zhang, X., Fei, N., Zhang, X., Wang, Q., & Fang, Z. (2022). Machine Learning Prediction Models for Postoperative Stroke in Elderly Patients: Analyses of the MIMIC Database. *Frontiers in Aging Neuroscience*, *14*, 897611. https://doi.org/10.3389/fnagi.2022.897611

DATA SOURCE

Dataset:

Training set: The Medical Information Mart for Intensive Care MIMIC-IV

Independent set: The Medical Information Mart for Intensive Care MIMIC-III

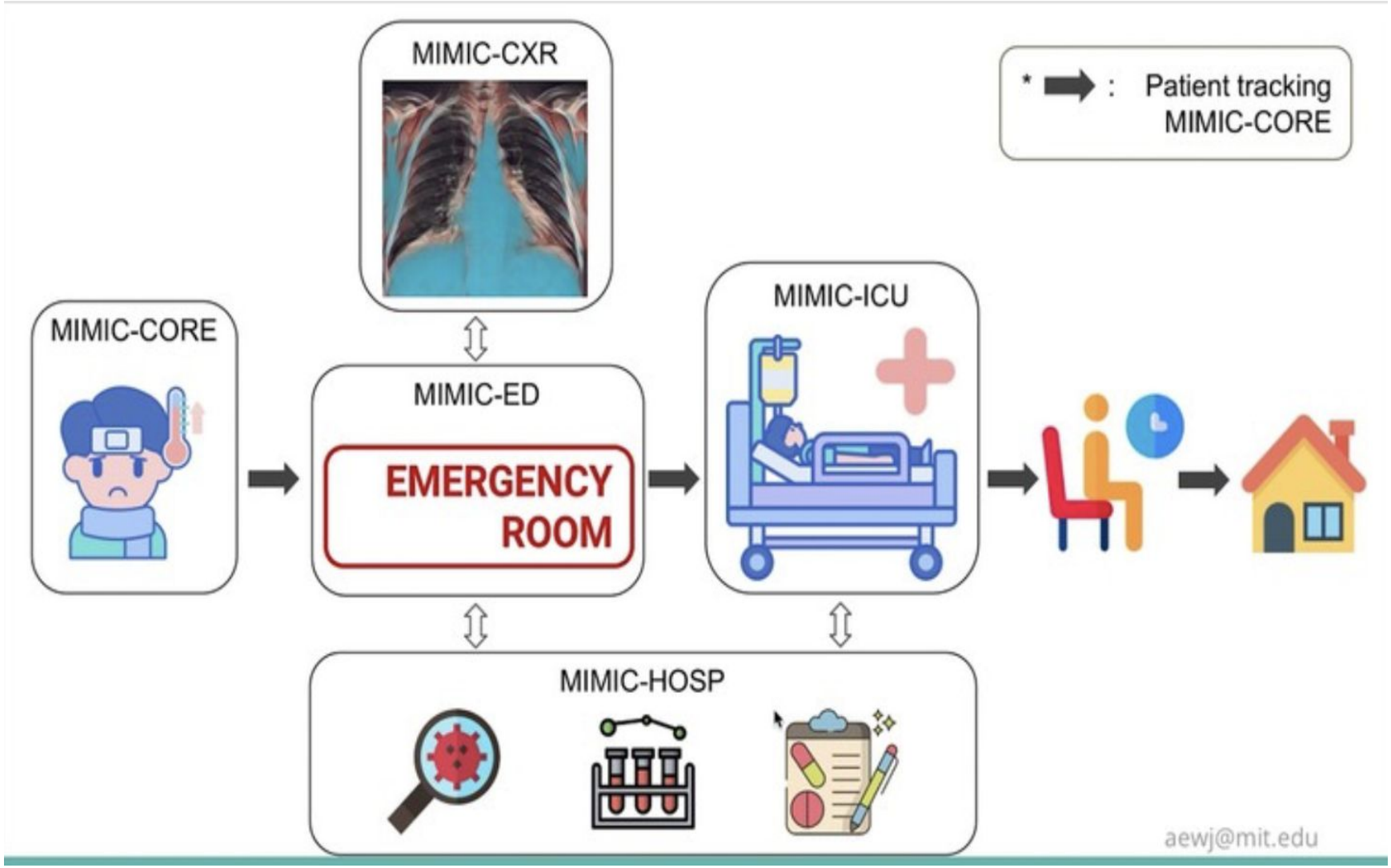
Tables used:

- PATIENTS: demographics, anchor age
- ADMISSIONS: hospital admissions
- ICUSTAYS: SICU admissions and stay times
- DIAGNOSES_ICD: ICD-coded diagnoses
- LAB EVENTS: Lab results
- CHART EVENTS: Vital signs in the ICU

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Data Source

MIMIC IV

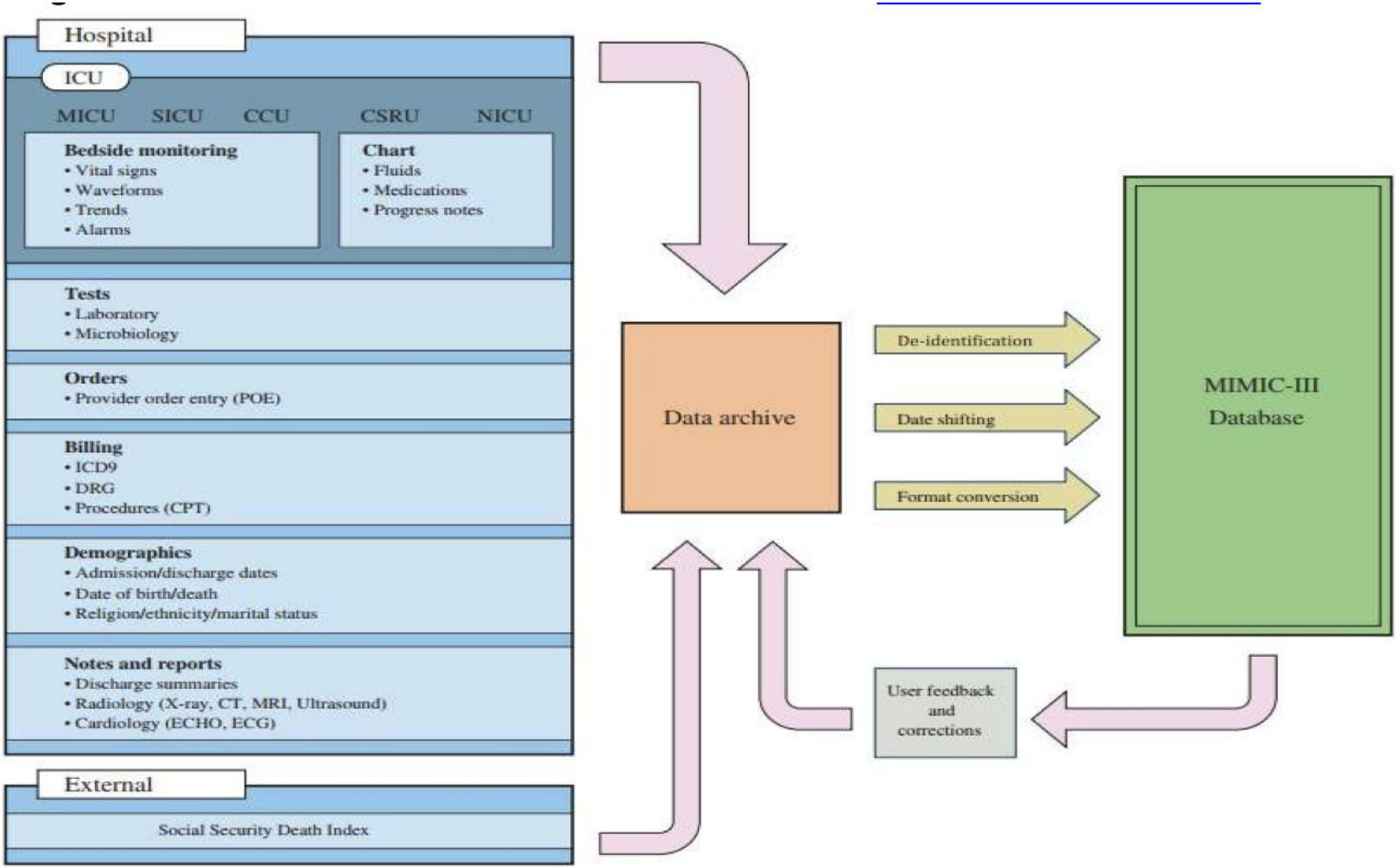


- Alistair Johnson @ Conference for Health, Inference and Learning (CHIL 2020)

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Data Source

MIMIC III

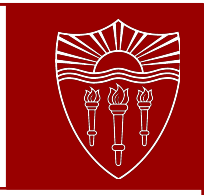


Johnson et al

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Data Extraction

MIMIC IV - Training set



- **Inclusion:** Patients from SICU department.
- **Exclusion:** First stay in SICU and Age < 55 years

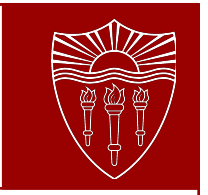
Work	Zhang et al. (2022)	Our work
Number of Patients	7128	7129
Stroke group	735	1355
Non-Stroke group	6393	5774

- Zhang, X., Fei, N., Zhang, X., Wang, Q., & Fang, Z. (2022). Machine Learning Prediction Models for Postoperative Stroke in Elderly Patients: Analyses of the MIMIC Database. *Frontiers in Aging Neuroscience*, *14*, 897611. https://doi.org/10.3389/fnagi.2022.897611

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Data Extraction

MIMIC III - Independent set



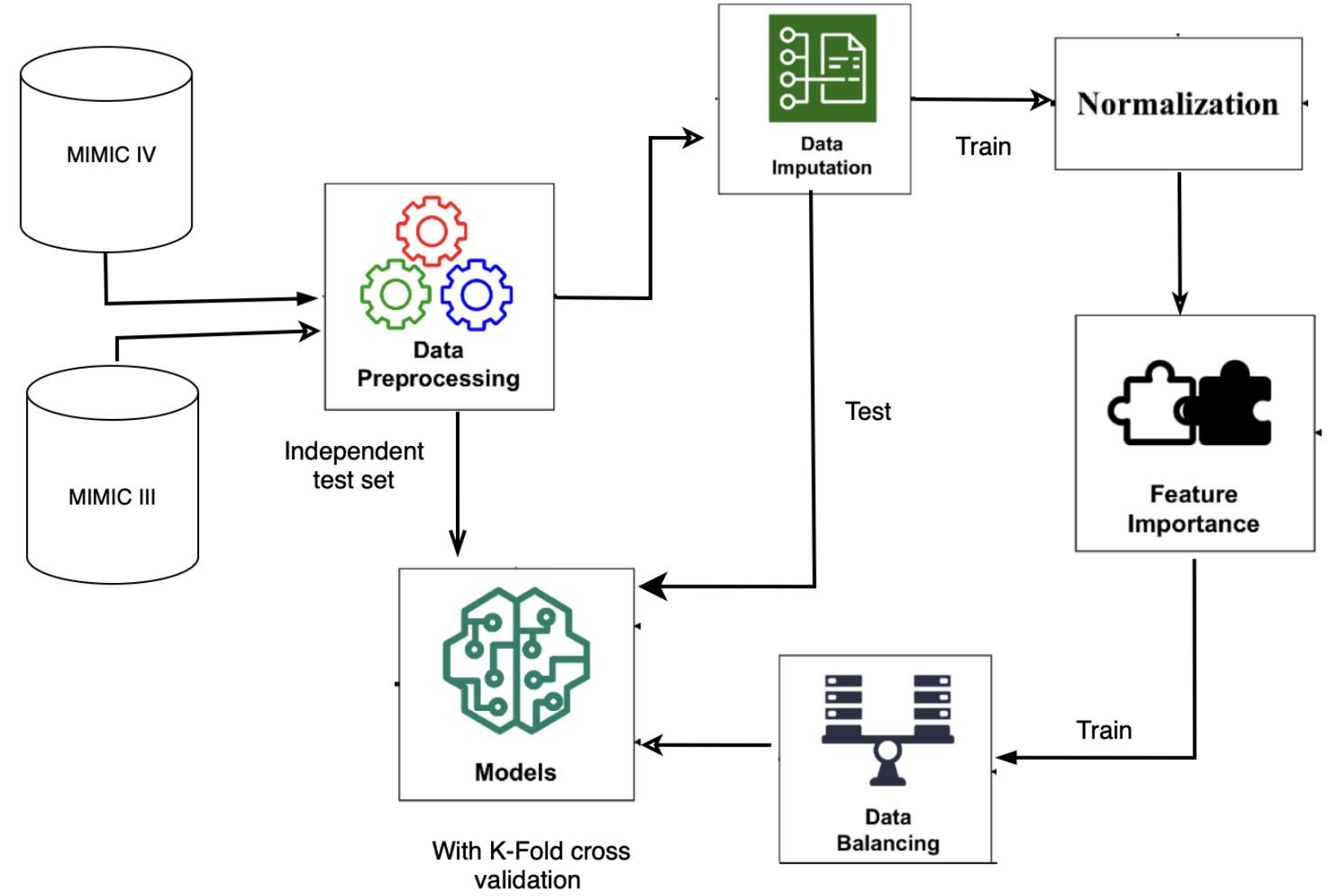
- **Inclusion:** Patients from SICU department.
- **Exclusion:** First stay in SICU, Age < 55 years and individual with data missing.

Work	Zhang et al. (2022)	Our work
Number of Patients	661	681
Stroke group	43	62
Non-Stroke group	618	619

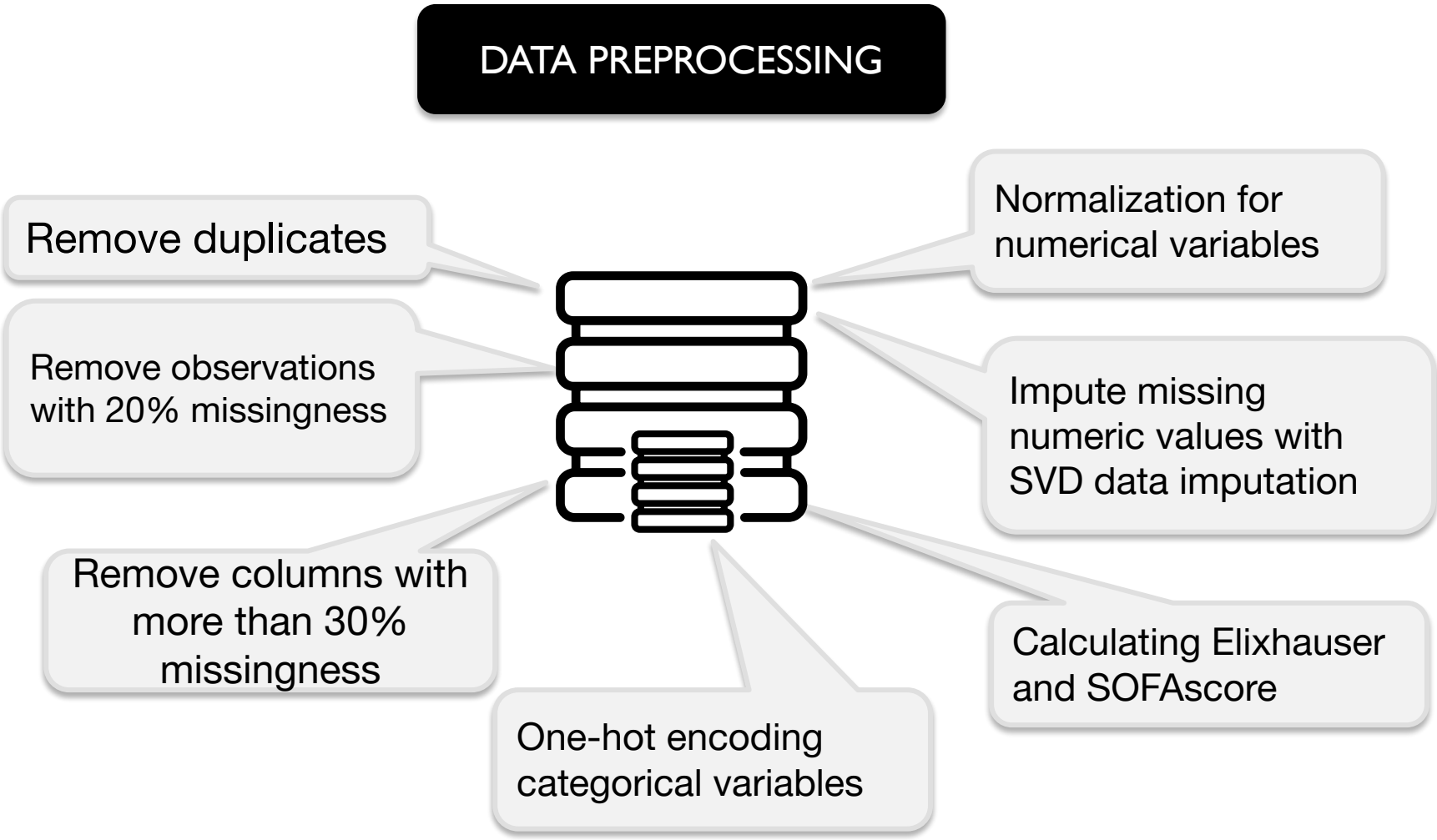
- Zhang, X., Fei, N., Zhang, X., Wang, Q., & Fang, Z. (2022). Machine Learning Prediction Models for Postoperative Stroke in Elderly Patients: Analyses of the MIMIC Database. *Frontiers in Aging Neuroscience*, *14*, 897611. https://doi.org/10.3389/fnagi.2022.897611

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Model pipeline



Data Preprocessing



Feature Selection



Features

Demographic:

- Age
- Sex
- Ethnicity
- insurance
- los

Comorbidity:

- Hypertension
- COPD
- PVD
- CHF
- Diabetes
- Renal
- Liver
- Sepsis
- Cancer
- Rheumatic
- Peptic ulcer disease
- first_care unit
- last_care unit
- Alcohol
- Tobacco
- Opioid
- Smoking
- Cocaine
- Cannabis
- Sedative
- Other

Lab results:

- anion gap
- albumin
- bilirubin
- creatinine
- hematocrit
- hemoglobin
- WBC
- lactate
- platelets
- K
- PTT
- PT
- INR
- Urea Nitrogen
- eGFR
- Sodium
- Calcium
- Magnesium
- Phosphate
- Chloride
- Ast
- CRP
- Troponin

Vitals signs:

- Heart Rate
- Systolic_BP
- Diastolic_BP
- Mean_BP
- Spo2

Pre-op medication:

- Anti-coagulants
- Aspirin
- Insulin
- diabetic
- statin

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Feature Selection



- Started with 108 features from the dataset.
- Applied RFECV with xgboost.
- Selected 80 features from the RFECV with rank 1.

- Applied permutation importance with Random forest on the 80 features.
- Selected top 25 features.

- Get SHAP based 30 features with Random Forest

- Retain top 20 features from the intersection of the both results

Feature Selection

Top 20 Features:

Category	Features
Demographics	age, los, gender
Comorbidities	Cancer, liver, chf, htn
Lab results	Calcium, Potassium, Platelet Count, PTT, WBC, Phosphate, chloride, Anion gap
Vital signs	systolic_BP, SpO2, Heart_Rate,
Scores	Elixhauser_score, SOFA_score

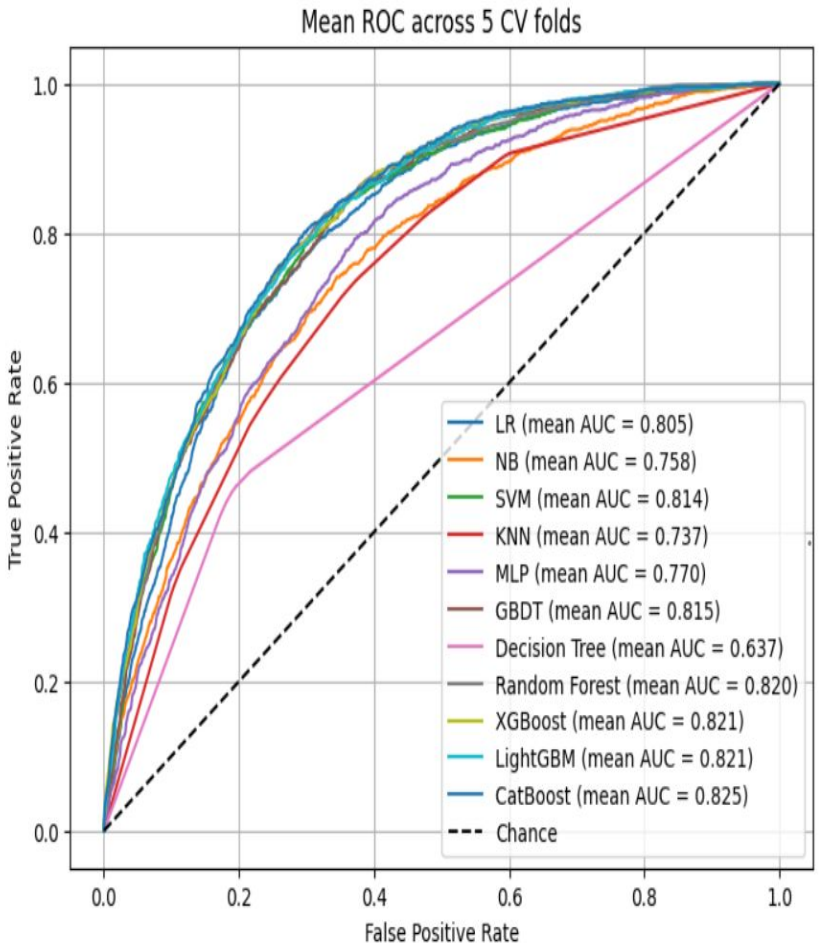
Data Modeling

- Machine Learning models are built with the resultant 20 features.
- MIMIC-IV is used for training the models with K fold cross validation and SMOTE for class imbalance and MIMIC-III is used for testing .

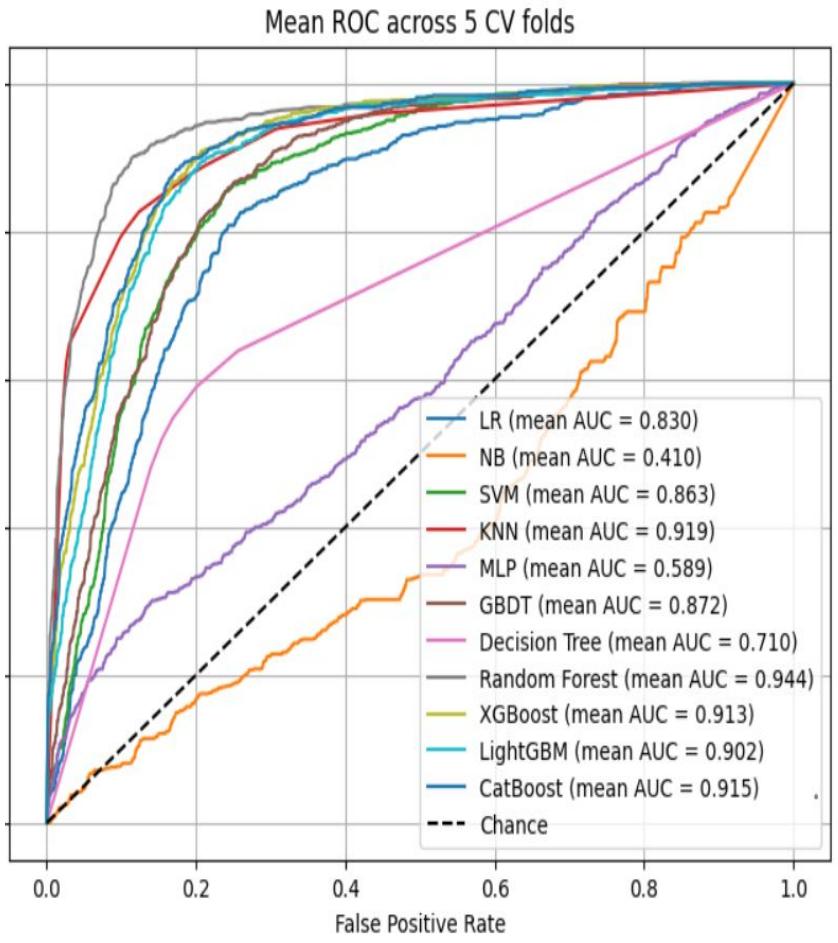
Machine Learning Models Used:

1. Logistic Regression
2. Decision Tree
3. Random Forest
4. K-Nearest Neighbors (KNN)
5. Support Vector Machine (SVM – RBF Kernel)
6. GBDT
7. Multi-Layer Perceptron (MLP)
8. XGBoost
9. CATBoost
10. LightGBM

Data Modeling



ROC Curve for training cohort



ROC curve for testing cohort

Data Modeling

```
AUC (5-fold CV) training set:
LR                  mean=0.805  std=0.006  folds=[0.801 0.807 0.797 0.814 0.806]
NB                  mean=0.758  std=0.010  folds=[0.757 0.747 0.762 0.773 0.749]
SVM                 mean=0.814  std=0.010  folds=[0.819 0.805 0.801 0.825 0.82 ]
KNN                 mean=0.738  std=0.014  folds=[0.723 0.736 0.731 0.761 0.739]
MLP                 mean=0.772  std=0.019  folds=[0.763 0.751 0.766 0.802 0.776]
GBDT                mean=0.816  std=0.007  folds=[0.805 0.813 0.82  0.82  0.822]
Decision Tree       mean=0.624  std=0.009  folds=[0.625 0.62  0.63  0.633 0.611]
Random Forest       mean=0.818  std=0.006  folds=[0.813 0.81  0.82  0.826 0.82 ]
XGBoost             mean=0.819  std=0.004  folds=[0.814 0.817 0.819 0.824 0.821]
LightGBM            mean=0.818  std=0.008  folds=[0.807 0.82  0.816 0.828 0.818]
CatBoost            mean=0.824  std=0.006  folds=[0.815 0.823 0.83  0.826 0.828]
-------------------------------------------------

AUC (5-fold CV) test set:
LR                  mean=0.829  std=0.004  folds=[0.826 0.825 0.834 0.831 0.831]
NB                  mean=0.416  std=0.002  folds=[0.415 0.418 0.415 0.415 0.418]
SVM                 mean=0.863  std=0.006  folds=[0.873 0.862 0.866 0.858 0.859]
KNN                 mean=0.918  std=0.010  folds=[0.915 0.926 0.928 0.902 0.92 ]
MLP                 mean=0.609  std=0.025  folds=[0.599 0.593 0.616 0.649 0.588]
GBDT                mean=0.873  std=0.005  folds=[0.875 0.867 0.876 0.868 0.878]
Decision Tree       mean=0.707  std=0.040  folds=[0.763 0.71  0.706 0.65  0.707]
Random Forest       mean=0.942  std=0.003  folds=[0.944 0.942 0.946 0.938 0.939]
XGBoost             mean=0.916  std=0.003  folds=[0.914 0.919 0.92  0.915 0.913]
LightGBM            mean=0.902  std=0.008  folds=[0.903 0.896 0.902 0.893 0.914]
CatBoost            mean=0.913  std=0.004  folds=[0.918 0.907 0.913 0.916 0.91 ]
```

- AUROC values are much better for testing set compared to the training set.
- RF, XGBoost, CATBoost, LightGBM produces very good results.

Results

- So we tuned the hyper parameters of all these models using GridSearchCV
- Training set performance comparison

Models	AUROC	Accuracy	F1-score	Sensitivity	Specificity
Random Forest	0.82 (0.81,0.83)	0.82 (0.81,0.83)	0.49 (0.47,0.51)	0.48 (0.45,0.51)	0.90 (0.88,0.91)
XGBoost	0.81 (0.80,0.83)	0.53 (0.52,0.54)	0.42 (0.41,0.44)	0.94 (0.92,0.95)	0.44 (0.43,0.45)
CATBoost	0.824 (0.81,0.84)	0.822 (0.81,0.83)	0.48 (0.45,0.5)	0.44 (0.42,0.48)	0.91 (0.90,0.93)
LightGBM	0.82 (0.81,0.84)	0.77 (0.76,0.78)	0.52 (0.50,0.54)	0.69 (0.67,0.71)	0.78 (0.77,0.80)

Results

Test set Performance comparison

Models	AUROC	Accuracy	F1-score	Sensitivity	Specificity
Random Forest	0.94 (0.933,0.941)	0.89 (0.87,0.90)	0.71 (0.69,0.74)	0.65 (0.61,0.68)	0.95 (0.94,0.96)
XGBoost	0.89 (0.87,0.90)	0.63 (0.61,0.65)	0.53 (0.51,0.55)	0.95 (0.93,0.97)	0.54 (0.52,0.56)
CATBoost	0.91 (0.89,0.92)	0.89 (0.86,0.91)	0.66 (0.62,0.70)	0.66 (0.62,0.70)	0.91 (0.90,0.92)
LightGBM	0.89 (0.87,0.90)	0.80 (0.79,0.81)	0.65 (0.63,0.68)	0.86 (0.83,0.88)	0.79 (0.77,0.80)

Results

Confusion matrices

	Random Forest	XGBoost	CATBoost	LightGBM
Training Data	[[5011 610] [652 603]]	[[2461 3160] [79 1176]]	[[5097 524] [699 556]]	[[4398 1223] [390 865]]
Test Data	[[2534 126] [266 479]]	[[1427 1233] [36 709]]	[[2410 250] [255 490]]	[[2082 578] [104 641]]

Proposed Model

Our proposed model is CATBoost

- As it demonstrated the best combination of discrimination in classes (AUROC=0.92 and 95% CI: (0.89,0.93)), sensitivity (0.66), specificity (0.91), and F1-score (0.66) on the independent test cohort.
- While also maintaining stable cross-validated performance during training.

Comparison	AUROC	Accuracy	Sensitivity	Specificity
Our work (CATBoost)	0.92 (0.89,0.93)	0.89 (0.87,0.91)	0.66 (0.62,0.70)	0.91 (0.90,0.92)
Zhang et al 2022 (XGBoost)	0.83 (0.79–0.87)	0.83 (0.79-0.87)	0.97 (0.96-0.98)	0.3 (0.19-0.45)

- Zhang, X., Fei, N., Zhang, X., Wang, Q., & Fang, Z. (2022). Machine Learning Prediction Models for Postoperative Stroke in Elderly Patients: Analyses of the MIMIC Database. *Frontiers in Aging Neuroscience*, *14*, 897611. https://doi.org/10.3389/fnagi.2022.897611

Proposed Model

Comparison of confusion matrices:

	Our Work (CATBoost)	Zhang et al (2022) (XGBoost)
Training set:	$\begin{bmatrix} 5097 & 524 \\ 699 & 556 \end{bmatrix}$	$\begin{bmatrix} 988 & 292 \\ 52 & 94 \end{bmatrix}$
Independent test set:	$\begin{bmatrix} 2410 & 250 \\ 255 & 490 \end{bmatrix}$	$\begin{bmatrix} 572 & 46 \\ 19 & 24 \end{bmatrix}$

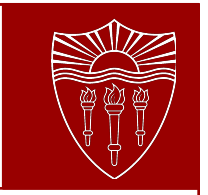
- Zhang, X., Fei, N., Zhang, X., Wang, Q., & Fang, Z. (2022). Machine Learning Prediction Models for Postoperative Stroke in Elderly Patients: Analyses of the MIMIC Database. *Frontiers in Aging Neuroscience*, *14*, 897611. https://doi.org/10.3389/fnagi.2022.897611

Proposed Model

- CATBoost model shows superior overall predictive quality compared to Zhang et al.'s XGBoost model as it has higher accuracy and AUROC.
- In sensitivity and sensibility trade-off, Zhang et al's XGBoost model is extremely aggressive at predicting stroke and almost every patient is flagged "high risk," causing high false positives.
- CATBoost learns a more calibrated decision boundary and it shows good discrimination between stroke vs. non-stroke.
- Compared to Zhang et al.'s 2022 XGBoost model, our CATBoost model achieves markedly higher AUROC, accuracy, and specificity, while maintaining reasonable sensitivity. The improvement in discriminative ability and reduction in false positives make our model more clinically applicable for real-world postoperative stroke prediction.

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Statistical tests

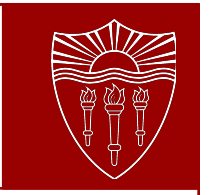


› Comparison of continuous variables between training and test sets using independent two-sample t-tests.

Feature	Training Set (Mean ± SD)	Test Set (Mean ± SD)	P-value
los	3.89 ± 5.52	6.03 ± 8.32	0.0
Alcohol	0.03 ± 0.16	0.00 ± 0.00	0.0
Tobacco	0.03 ± 0.17	0.00 ± 0.00	0.0
sepsis	0.14 ± 0.34	0.22 ± 0.42	0.0
cancer	0.19 ± 0.39	0.37 ± 0.48	0.0
mean_Urea Nitrogen	24.43 ± 18.72	30.09 ± 22.32	0.0
mean_chloride	104.12 ± 5.49	105.42 ± 6.03	0.0
Opioid	0.00 ± 0.07	0.00 ± 0.00	0.0
min_Urea Nitrogen	23.14 ± 17.78	28.47 ± 21.21	0.0
max_Urea Nitrogen	25.71 ± 19.94	31.70 ± 23.63	0.0

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Statistical tests

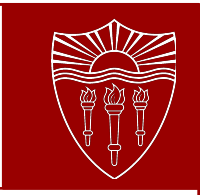


› Comparison of Numerical Feature Distributions in Stroke and Non-Stroke Patient Groups via T-Test

Variable	Non-Stroke (Mean ± SD)	Stroke (Mean ± SD)	p-value
max_diastolic_BP	111.98 ± 1116.20	109.85 ± 350.17	0.9098
max_Heart Rate	101.73 ± 21.96	102.31 ± 175.09	0.9068
mean_Anion Gap	14.49 ± 3.67	14.51 ± 2.63	0.7956
max_Spo2	101.24 ± 133.51	100.11 ± 25.31	0.5566
max_mean_BP	100.29 ± 108.29	209.56 ± 3503.03	0.2854
min_Mg	1.87 ± 0.33	1.88 ± 0.26	0.1864
Cocaine	0.00 ± 0.05	0.00 ± 0.03	0.1382
max_Platelet Count	223.70 ± 108.89	227.97 ± 86.96	0.1358
Sedative	0.00 ± 0.02	0.00 ± 0.00	0.0833
max_systolic_BP	150.94 ± 310.72	158.91 ± 23.68	0.0726

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Statistical tests



› Comparison of Categorical Feature Distributions in Stroke and Non-Stroke Patient Groups via Chi-square Test

Variable	Non-Stroke (%)	Stroke (%)	p-value
first_careunit (Surgical Intensive Care Unit (...	100.0%	100.0%	1.0000
insurance (Medicare)	52.4%	54.6%	0.0719
insurance (Medicaid)	5.1%	3.7%	0.0719
insurance (Other)	42.5%	41.8%	0.0719
gender (F)	46.9%	51.2%	0.0067
gender (M)	53.1%	48.8%	0.0067
last_careunit (Cardiac Vascular Intensive Care...	0.3%	0.0%	0.0000
last_careunit (Neuro Stepdown)	0.2%	0.8%	0.0000
last_careunit (Neuro Intermediate)	0.2%	2.4%	0.0000
last_careunit (Medical/Surgical Intensive Care...	0.7%	0.0%	0.0000

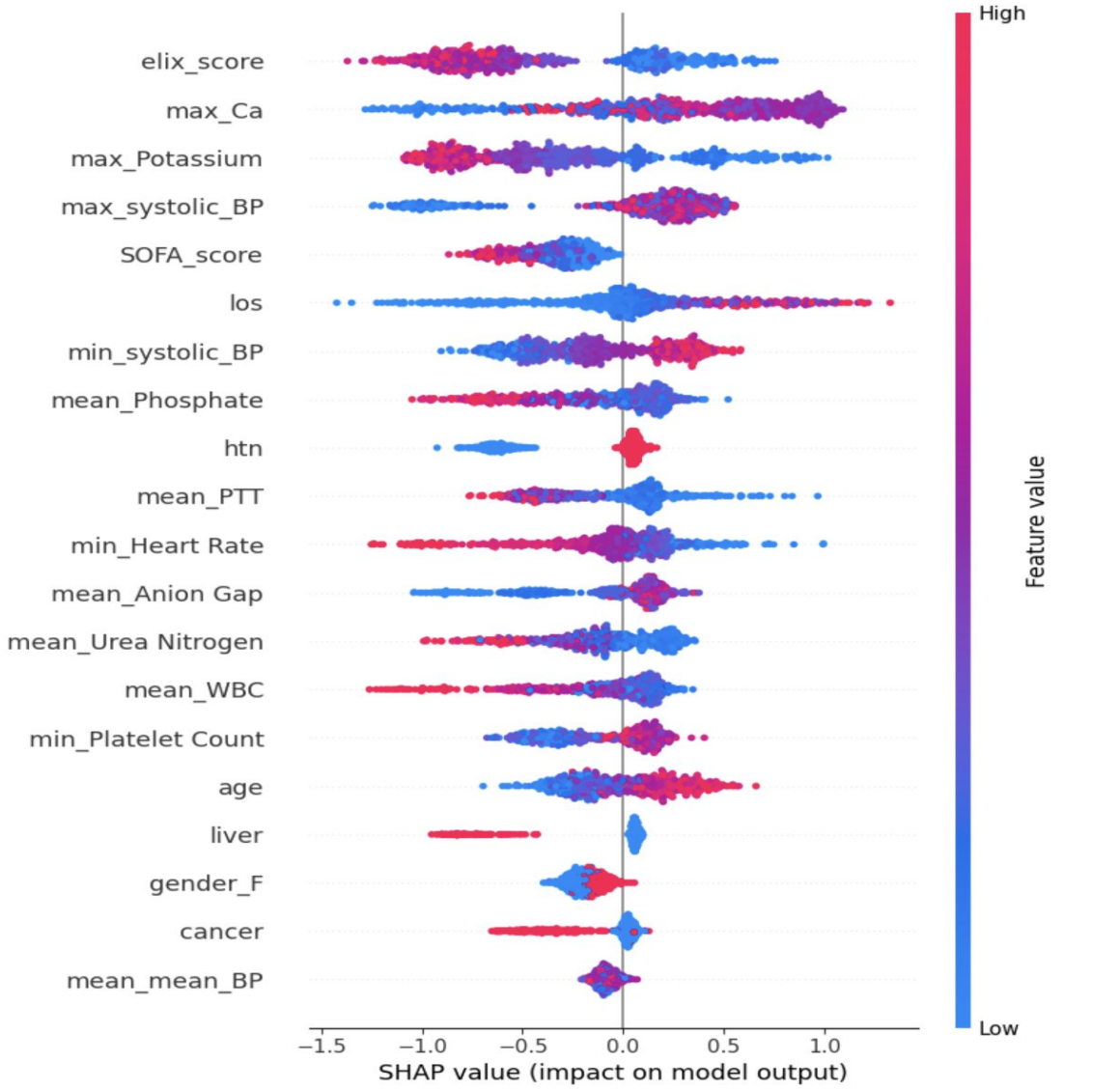
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Model Interpretation

SHAP Analysis

SHAP Analysis:

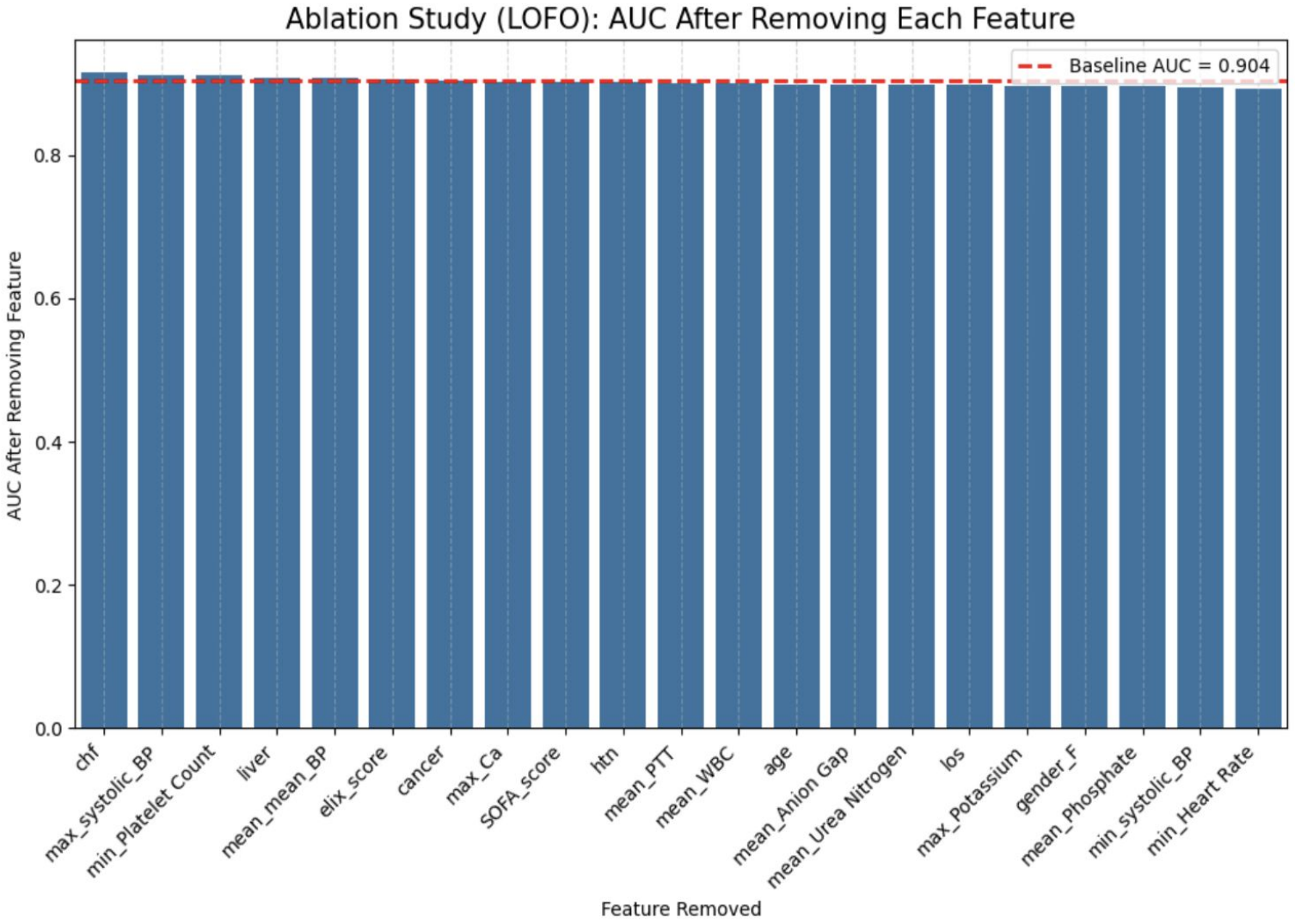
- The strongest predictors are the Elixhauser score, max_calcium and max_potassium, blood pressure extremes, and the SOFA score. For these features higher the feature value increases model-predicted stroke risk.



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Model Interpretation

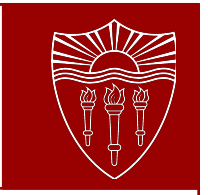
Ablation Study



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Model Interpretation

Ablation Study



- Ablation analysis demonstrated that removing any individual feature resulted in minimal changes to model performance, with AUC values remaining essentially unchanged from the baseline AUC of 0.904.
- This finding indicates that no single feature is indispensable for prediction; rather, the CatBoost model leverages a distributed set of physiological and comorbidity-related variables to estimate postoperative stroke risk.
- The robustness of the model to single-feature removal suggests redundancy among features and reflects the multifactorial nature of stroke risk in elderly postoperative patients.
- But removing features doesn't affect performance, it suggests that the model relies more on interaction effects than single features and the predictive signal is distributed, making clinical interpretation harder

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Conclusion

- Through a comprehensive preprocessing and modeling pipeline **CatBoost** was identified as the top-performing algorithm. It achieved an **AUROC of 0.92 (95% CI: 0.89–0.93)**, outperforming other gradient boosting frameworks such as XGBoost and LightGBM.

- The independent external test cohort (MIMIC-III) exhibited higher AUROC values compared to internal cross-validation on the MIMIC-IV training data. This suggests that MIMIC-III patients had more homogeneous or more clearly separable postoperative physiological patterns, leading to improved model performance. The results indicate strong generalizability rather than overfitting.

- The model's reliance on meaningful predictors (SOFA, Elixhauser index, electrolytes, BP extremes, inflammatory markers) confirms that it is learning clinically coherent patterns rather than noise.

- The ablation study showed that removing individual features did not significantly reduce AUC, indicating that the model is highly **robust**, with predictive information distributed across multiple correlated clinical variables.

Citations

- Li et al, Predicting Postoperative Stroke in Elderly SICU Patients: An Interpretable Machine Learning Model Using MIMIC Data, 2025, https://arxiv.org/abs/2506.03209.

- Zhang, X., Fei, N., Zhang, X., Wang, Q., & Fang, Z. (2022). Machine Learning Prediction Models for Postoperative Stroke in Elderly Patients: Analyses of the MIMIC Database. *Frontiers in Aging Neuroscience*, *14*, 897611. https://doi.org/10.3389/fnagi.2022.897611

- Alistair Johnson @ Conference for Health, Inference and Learning (CHIL 2020)



THANK YOU